AS · Unit 2 · WCH12 | Energetics, Group Chemistry, Halogenoalkanes and Alcohols

Halogenoalkanes, alcohols and spectra

Swap one hydrogen in an alkane for a halogen or an –OH group and the molecule suddenly has a polar bond that other species can attack. This topic follows what happens next: how halogenoalkanes are substituted or lose HX to give alkenes, how alcohols are oxidised, halogenated and dehydrated, how the products are separated and purified in the lab, and how mass spectra and infrared spectra tell you which molecule you have made.

SPEC 10.1 Classifying reactions

WHAT THIS SPEC POINT ASKS

Classify organic reactions (including the Unit 1 ones) as addition, elimination, substitution, oxidation, reduction, hydrolysis or polymerisation.

The seven types

Type	What happens	Examples
Addition	two molecules join to make **one** product; a double bond opens	ethene + Br_2 → 1,2-dibromoethane; ethene + HBr; ethene + H_2O (steam)
Elimination	a small molecule (H_2O, HX) is **removed** from one molecule, leaving a **C=C**	halogenoalkane + ethanolic KOH → alkene; alcohol + conc. H_3PO_4 → alkene
Substitution	one atom or group is **swapped** for another	alkane + Cl_2 (UV) → chloroalkane + HCl; halogenoalkane + OH^- → alcohol + X^-
Oxidation	gain of O / loss of H (or electrons)	alcohol + acidified $K_2Cr_2O_7$; combustion; alkene + acidified $KMnO_4$ → diol
Reduction	gain of H / loss of O	alkene + H_2 (Ni catalyst) → alkane
Hydrolysis	a bond is **split by water** (or OH^-)	halogenoalkane + H_2O or OH^- → alcohol
Polymerisation	many small monomers join into a long chain	ethene → poly(ethene) (addition polymerisation)

- One reaction can fit more than one label:

    - halogenoalkane + OH^- → alcohol is **substitution** and **hydrolysis**;
    - alkene + H_2 is **addition** and **reduction**;
    - alcohol + conc. H_3PO_4 is **elimination**, also called **dehydration**.
- Answer with the label the question is after. If it asks for the *type of reaction*, one word (e.g. "elimination") is enough; if it asks for the *mechanism*, add the attacking species: "**nucleophilic** substitution", "**electrophilic** addition", "**free-radical** substitution".

MARK-SCHEME WORDING

- A table of reagent, solvent and type (WCH12 Oct 2020): KOH(aq) → **substitution**; NH_3 in ethanol → substitution; KCN in ethanol → substitution; **KOH in ethanol → elimination**. "Nucleophilic" is ignored for the substitutions; "electrophilic" or "free radical" is penalised.
- "**elimination**": DO NOT AWARD "addition–elimination" or "nucleophilic elimination" (WCH12 Jan 2025, Jun 2024).
- IGNORE "dehydration" for elimination and "displacement" for substitution; DO NOT AWARD "dehydrogenation" (WCH12 Jun 2022).

Often asked as: name the type of reaction for a given equation or reagent, often in a reaction-scheme question.

COMMON MISTAKE

Calling the halogenoalkane + **ethanolic** KOH reaction "nucleophilic substitution". It's **elimination** (the most common wrong answer, WCH12 Jan 2025 report).

Giving a mechanism when the question asked only for the type.

SPEC 10.2 Reaction mechanisms

WHAT THIS SPEC POINT ASKS

Understand what a reaction mechanism is.

Equation v mechanism

- An **equation** shows only the starting materials and the products.
- A **mechanism** shows **how** the reaction happens, step by step: which bonds break, which form, and in what order, using **curly arrows** to show where electrons go.
- Mechanisms explain why reactions follow patterns: reagents that react the same way follow the same mechanism (all primary halogenoalkanes with OH^-, for example).

Curly arrows

- A **full-headed curly arrow** shows the movement of a **pair of electrons**. These are the arrows in every Topic 10 mechanism.
- A **half-headed arrow (fish-hook)** shows the movement of **one electron**, as in free-radical reactions (Unit 1).
- Rules:
  - an arrow **starts** at a **lone pair** or at the **middle of a covalent bond** (where the electrons are), never at a charge or an atom's symbol on its own;
  - it **ends** at the atom the electrons go to (to form a new bond) or just past an atom (when it leaves as an ion).
- Show **partial charges** (δ+ / δ−) on polar bonds and **full charges** on ions.

MARK-SCHEME WORDING

- "The curly arrow should start from a **covalent bond or a lone pair**" (WCH12 Jun 2023 report). Half-arrows are penalised once.
- An arrow starting from the **negative charge** of OH^- instead of a lone pair loses the mark (WCH12 Oct 2021).

Often asked as: draw a mechanism; spot what's missing from a given mechanism ("identify the omissions", WCH12 Jan 2022).

COMMON MISTAKE

Arrows drawn to show where an **atom** moves, not where the electron pair moves (e.g. an arrow from H to O, when it should go from O's lone pair to H).

Fish-hook arrows in a nucleophilic substitution.

SPEC 10.3 Heterolytic bond breaking: electrophiles and nucleophiles

WHAT THIS SPEC POINT ASKS

Understand that heterolytic bond breaking produces species that act as electrophiles or nucleophiles.

Two ways a covalent bond can break

	Homolytic fission	Heterolytic fission
Electrons	**one** electron to each atom	**both** electrons to **one** atom
Products	two **radicals** (unpaired electron, e.g. Cl•)	a **positive ion** and a **negative ion**
Arrows	half-headed	full-headed
Typical when	non-polar bond (Cl–Cl, C–H) with UV light	polar bond (C–Br, H–Br)
Leads to	free-radical reactions	electrophilic and nucleophilic reactions

- In heterolytic fission the electrons go to the **more electronegative** atom. For C–Br: CH_3CH_2–Br → $CH_3CH_2^+$ + Br^-.
- The ions formed are reactive:
  - the **positive** species is electron-poor: it can **accept** a pair of electrons, so it's an **electrophile** (e.g. a carbocation, H^+, NO_2^+);
  - the **negative** species has a spare lone pair: it can **donate** a pair of electrons, so it's a **nucleophile** (e.g. Br^-, OH^-).
- A **carbocation** is an ion with a positive charge on a carbon atom. It has three bonding pairs and no lone pairs around the C^+, so it is **trigonal planar** (bond angle 120°).

MARK-SCHEME WORDING

- "**heterolytic** (bond) fission" and "the (pair of) electrons in the C–Br bond are transferred to the **bromine** atom". There must be a mention of **two electrons** (WCH12 Jun 2023).
- Carbocation shape: "**trigonal planar**" because "three bond pairs (and no lone pairs) arranged to minimise repulsion" (WCH12 Jun 2023).

Often asked as: name the type of bond breaking in the first step of a mechanism; give the shape of a carbocation intermediate.

COMMON MISTAKE

Saying "the bond breaks and bromine takes an electron": it takes **both** electrons of the bond.

Stopping the shape explanation at "three bonded pairs". Say they **repel equally** and get as far apart as possible, giving 120°.

SPEC 10.4 Nucleophiles

WHAT THIS SPEC POINT ASKS

Know the definition of a nucleophile.

Definition

- A **nucleophile** is a species that **donates a lone pair of electrons** to an electron-deficient (δ+ or +) atom, forming a new **covalent bond**.
- "Nucleophile" means "nucleus-loving": it is attracted to positive centres, such as the **δ+ carbon** in a C–X bond.
- Every nucleophile has a **lone pair**. Many are negative ions, but some are neutral.

Nucleophile	Where the lone pair is	Product with a halogenoalkane
OH^-	on O	alcohol
CN^-	on C	nitrile
NH_3	on N (neutral molecule)	amine
H_2O	on O (neutral, weak)	alcohol
RNH_2 (an amine)	on N	secondary amine (unwanted side reaction)

- The opposite is an **electrophile**: a species that **accepts a pair of electrons** (e.g. H^+, $Br^{\delta+}$ in a polarised Br_2, a carbocation). Electrophiles attack electron-rich areas such as C=C (Unit 1).

MARK-SCHEME WORDING

- "**electron pair donor**", or "(species with a) **lone pair** that attacks the δ+ carbon" (WCH12 Jan 2022, Jan 2026). "Acts as a nucleophile" is accepted for the role of OH^- in substitution; just repeating "nucleophilic substitution" is ignored.
- An amine is a nucleophile because of the "**lone pair on the nitrogen**"; it can be substituted further because it still has "**H atoms attached to N**" (WCH12 Jun 2025).

Often asked as: define a nucleophile; state the role of OH^- in a given reaction (nucleophile in substitution, **base** in elimination).

COMMON MISTAKE

"A nucleophile is a negative ion." NH_3 and H_2O are neutral nucleophiles. The key is the **lone pair**.

"Electron donor" without "**pair**".

SPEC 10.5 Bond polarity and the type of mechanism

WHAT THIS SPEC POINT ASKS

Understand how bond polarity decides the type of mechanism a compound undergoes.

Match the bond to the attacker

Compound	Key bond	Polarity	Attacked by	Mechanism
Alkane	C–H, C–C	non-polar	radicals (Cl• from UV light)	**free-radical substitution** (homolytic fission)
Alkene	C=C	non-polar but a region of **high electron density** (the π bond)	**electrophiles** (Br_2, HBr)	**electrophilic addition**
Halogenoalkane	$C^{\delta+}–X^{\delta-}$	**polar**	**nucleophiles** (OH^-, CN^-, NH_3)	**nucleophilic substitution** (heterolytic fission)

- Halogens are **more electronegative** than carbon, so the C–X bonding pair is pulled towards X. The carbon becomes **δ+**, the halogen **δ−**.
- The δ+ carbon attracts nucleophiles. The C–X bond then breaks **heterolytically**, and the halogen leaves as a halide ion, X^-.
- So **the polarity of a bond tells you where a reagent will attack and whether the bond breaks evenly or unevenly**.

EXAM TIP

Electronegativity falls down Group 7 (F > Cl > Br > I), so C–F is the **most** polar bond and C–I the **least**. If polarity decided the rate, fluoroalkanes would react fastest. They don't: **bond enthalpy** decides the rate of hydrolysis (10.13).

MARK-SCHEME WORDING

- Dipoles: "**δ+ on C, δ− on the halogen**", shown on the C–X bond in every substitution mechanism.

Often asked as: part of a mechanism question; explain why a halogenoalkane reacts with nucleophiles but an alkane doesn't.

COMMON MISTAKE

Putting the δ+ on the halogen. Writing full + and − charges on a polar bond in the starting molecule.

SPEC 10.6 Naming and drawing halogenoalkanes

WHAT THIS SPEC POINT ASKS

Name halogenoalkanes and draw their structural, displayed and skeletal formulae.

Naming rules

1. Find the **longest carbon chain** containing the C–X carbon: it gives the stem (meth-, eth-, prop-, but-, pent-, hex-) + **-ane**.
2. Halogens are **prefixes**: **fluoro-, chloro-, bromo-, iodo-**. Alkyl branches are prefixes too (methyl-, ethyl-).
3. Number the chain to give the **lowest numbers** to the substituents.
4. Put all prefixes in **alphabetical order** (bromo, chloro, ethyl, fluoro, iodo, methyl). Use di-, tri- for repeats (they don't count for alphabetical order).
5. Commas between numbers, hyphens between numbers and words.

Name	Structural formula	Class (10.7)
bromoethane	CH_3CH_2Br	primary
1-bromopropane	$CH_3CH_2CH_2Br$	primary
2-bromopropane	$CH_3CHBrCH_3$	secondary
1-bromo-2-methylpropane	$(CH_3)_2CHCH_2Br$	primary
2-chloro-2-methylpropane	$(CH_3)_3CCl$	tertiary
2-iodo-2-methylpropane	$(CH_3)_3CI$	tertiary
2-chloro-3-methylbutane	$CH_3CHClCH(CH_3)CH_3$	secondary
1,2-dibromoethane	CH_2BrCH_2Br	primary (both)

The three formula types

- **Structural**: enough to show the arrangement without drawing bonds, e.g. $CH_3CHBrCH_3$.
- **Displayed**: every atom and **every bond** drawn (including all C–H bonds).
- **Skeletal**: a zig-zag line; every corner and line end is a carbon; H atoms on carbon are **not shown**; the halogen is written at the end of its bond. 2-bromopropane is a "V" shape (three carbons) with Br on a line from the middle carbon.

MARK-SCHEME WORDING

- "2-chloro-3-methylbutane" (WCH12 Jan 2025). Missing or extra hyphens and commas are ignored; wrong alphabetical order isn't credited.
- "2-iodo-2-methylpropane": DO NOT AWARD "2-iodine-…" (WCH12 Oct 2024).
- Nitrile names count the C of the C≡N in the chain: $(CH_3)_3CCN$ is **2,2-dimethylpropanenitrile** (WCH12 Jun 2023).

Often asked as: name a compound from its skeletal formula; draw a displayed or skeletal formula from a name.

COMMON MISTAKE

Writing "chloride", "chlorine" or "iodine" in the name instead of chloro- / iodo-.

Prefixes out of alphabetical order (methyl before chloro).

Missing H atoms in displayed formulae; drawing 2-bromopropane when 1-bromo-2-methylpropane was asked (WCH12 Jan 2024 report).

SPEC 10.7 Primary, secondary and tertiary halogenoalkanes

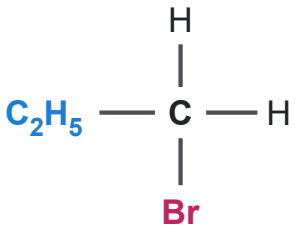
WHAT THIS SPEC POINT ASKS

Distinguish primary, secondary and tertiary halogenoalkanes.

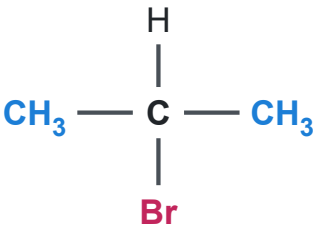
Count the carbons on the C–X carbon

Look at the carbon that carries the halogen. Count how many **other carbon atoms (alkyl groups)** are bonded to it:

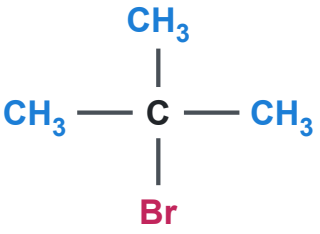
Class	Carbons bonded to the C–X carbon	General	Example
Primary (1°)	one (or none, for CH_3X)	RCH_2X	1-bromobutane
Secondary (2°)	two	R_2CHX	2-bromobutane
Tertiary (3°)	three	R_3CX	2-bromo-2-methylpropane



1-bromopropane
primary
1 carbon attached



2-bromopropane
secondary
2 carbons attached



2-bromo-2-methylpropane
tertiary
3 carbons attached

Count the carbon groups (blue) on the carbon that carries the halogen: one for primary, two for secondary, three for tertiary.

- The four structural isomers of C_4H_9Br:
  - 1-bromobutane $CH_3CH_2CH_2CH_2Br$: **primary**;
  - 2-bromobutane $CH_3CHBrCH_2CH_3$: **secondary**;
  - 1-bromo-2-methylpropane $(CH_3)_2CHCH_2Br$: **primary** (the Br carbon has only one C attached, even though the chain is branched);
  - 2-bromo-2-methylpropane $(CH_3)_3CBr$: **tertiary**.
- The class controls how fast the halogenoalkane is hydrolysed (10.12).

MARK-SCHEME WORDING

- Define by the number of **carbon atoms / alkyl groups attached to the carbon bonded to the halogen**. "Branched" is accepted for tertiary in rate questions (WCH12 Jun 2024).

Often asked as: classify a given halogenoalkane; draw isomers of a formula and classify each.

COMMON MISTAKE

Classifying by "how many halogens" or "where the halogen is on the chain".

Calling bromocyclohexane tertiary (it's **secondary**: the Br carbon has two C neighbours) (WCH12 Jun 2022 report).

SPEC 10.8 Reactions of halogenoalkanes

WHAT THIS SPEC POINT ASKS

Know the reactions of halogenoalkanes with (i) aqueous KOH (alcohols; OH^- as nucleophile), (ii) ethanolic KOH (alkenes by elimination; OH^- as base), (iii) aqueous silver nitrate in ethanol (water as nucleophile), (iv) ammonia in ethanol under pressure (amines), (v) KCN in ethanol (nitriles: lengthens the carbon chain).

Summary table

Reagent and conditions	Role of reagent	Type	Product (from bromoethane, or as stated)
KOH(aq) or NaOH(aq), **heat under reflux**	OH^- = **nucleophile**	nucleophilic substitution (hydrolysis)	**ethanol**, CH_3CH_2OH
KOH in ethanol (ethanolic), heat under reflux	OH^- = **base**	**elimination**	**alkene** (from 2-bromopropane: propene)
$AgNO_3$(aq) in ethanol, warm	H_2O = **nucleophile**	nucleophilic substitution (hydrolysis)	alcohol + silver halide precipitate
excess NH_3 in ethanol, heat under pressure (sealed tube)	NH_3 = **nucleophile**	nucleophilic substitution	**amine**: ethylamine $CH_3CH_2NH_2$
KCN in ethanol, heat under reflux	CN^- = **nucleophile**	nucleophilic substitution	**nitrile**: propanenitrile CH_3CH_2CN

(i) Aqueous alkali → alcohol

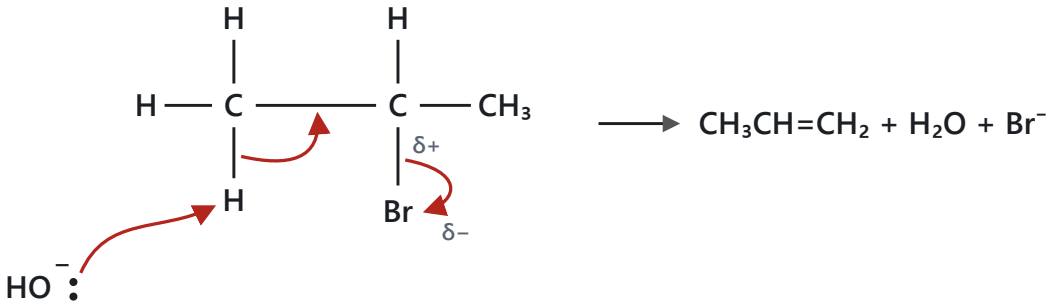
$CH_3CH_2Br + KOH \rightarrow CH_3CH_2OH + KBr$
Ionic: $CH_3CH_2Br + OH^- \rightarrow CH_3CH_2OH + Br^-$

- **Water** is the solvent, so OH^- ions are surrounded by water and attack the δ+ carbon. A little ethanol is often added to help the halogenoalkane mix with the water.

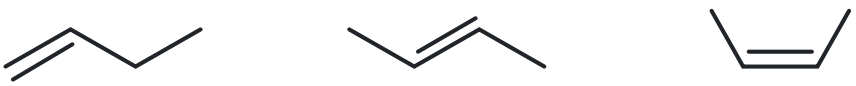
(ii) Ethanolic KOH → alkene (elimination)

$CH_3CHBrCH_3 + KOH \rightarrow CH_2{=}CHCH_3 + KBr + H_2O$

- In **ethanol** (no water), OH^- acts as a **base**: it removes an **H^+** from a carbon **next to** the C–X carbon. H and X are lost from **neighbouring** carbons, and a **C=C** forms between them.
- The ethanol must be **dry**: any water would let OH^- act as a nucleophile and give the alcohol instead.
- If the C–X carbon has different neighbouring carbons with H atoms, you get **more than one alkene**. 2-bromobutane gives **but-1-ene** and **but-2-ene** (which exists as **E** and **Z** isomers): three alkenes in all.
- In practice substitution and elimination compete. Aqueous conditions and primary halogenoalkanes favour substitution; ethanolic conditions, heat and tertiary halogenoalkanes favour elimination.



Elimination from 2-bromopropane. Three arrows: lone pair on O → H; C–H bond → C–C (forming C=C); C–Br bond → Br.



but-1-ene (E)-but-2-ene (Z)-but-2-ene

Removing H from carbon 1 gives but-1-ene; removing it from carbon 3 gives but-2-ene, which exists as E and Z isomers.

(iii) Silver nitrate in ethanol → alcohol + precipitate

$R{-}X + H_2O \rightarrow R{-}OH + H^+ + X^-$ then $Ag^+(aq) + X^-(aq) \rightarrow AgX(s)$

- Water is a **weak nucleophile**, so this hydrolysis is slow enough to time. The halide ion released is caught by Ag^+ as a precipitate: **AgCl white, AgBr cream, AgI yellow**.

- **Ethanol** is a solvent for both the halogenoalkane (which doesn't mix with water) and the aqueous silver nitrate, so they mix and react.
- This is the basis of the rate comparisons in 10.10 and Core Practical 5.

(iv) Ammonia → amine

$CH_3CH_2Br + 2NH_3 \rightarrow CH_3CH_2NH_2 + NH_4Br$

- Conditions: **excess** concentrated ammonia **in ethanol**, **heated under pressure** in a **sealed tube**. Reflux isn't used because ammonia gas would **escape through the condenser**.
- The second NH_3 removes H^+ from the intermediate (mechanism in 10.9), which is why the equation uses $2NH_3$.
- **Why excess ammonia?** The amine product also has a **lone pair on N**, so it is a nucleophile too and can attack more halogenoalkane, giving secondary and tertiary amines. A large excess of NH_3 makes it much more likely that each halogenoalkane molecule meets NH_3, not the amine.

(v) Potassium cyanide → nitrile

$CH_3CH_2Br + KCN \rightarrow CH_3CH_2CN + KBr$

- Conditions: **KCN in ethanol**, heat under reflux.
- The product, **propanenitrile**, has **three** carbons; bromoethane had two. Making a new **C–C bond** is a way of **increasing the length of the carbon chain**.
- Name nitriles by counting the C of –C≡N in the chain: CH_3CH_2CN is propane**nitrile**.
- KCN is **highly toxic**, and acids release toxic HCN gas.

MARK-SCHEME WORDING

- Substitution: "**aqueous** (solution)" (ALLOW aqueous ethanol); DO NOT AWARD "just water" (WCH12 Jun 2024).
- Elimination: "**ethanolic / alcoholic** (solution)" of KOH (ALLOW NaOH); DO NOT AWARD just "ethanol" or "alcohol" as the reagent. Heat/reflux ignored.
- Amine: "ammonia (ALLOW concentrated), **heat under pressure**, in **alcoholic solution**" or "in a **sealed tube**". DO NOT AWARD "heat under reflux" or "dilute / aqueous ammonia" (WCH12 Jun 2023, Jun 2025, Oct 2025).
- Nitrile: "$C_4H_9Cl + KCN \rightarrow C_4H_9CN + KCl$", "**pentanenitrile**", "a way of **extending the carbon chain** / forming a new C–C bond" (WCH12 Oct 2024, Oct 2025).
- Dry ethanol: "to remove water" so as to "prevent (nucleophilic) **substitution**" / stop OH^- acting as a nucleophile (WCH12 Oct 2025).
- 6-mark compare (WCH12 Jun 2022, Jun 2025): both involve OH^-; **aqueous** → **substitution**, OH^- is a **nucleophile**, alcohol formed; **ethanolic** → **elimination**, OH^- is a **base**, alkene(s) formed; two balanced equations. From 2-chlorobutane: butan-2-ol, and but-1-ene plus E- and Z-but-2-ene. Just "butene" scores only 1.

Often asked as: give reagents and conditions for a step in a reaction scheme; 6-mark comparison of aqueous and ethanolic KOH; state the role of OH^-; name the nitrile.

⚠ Beyond the spec · not part of the Edexcel IAL specification

Examined in past papers even though the spec doesn't list it.

- **Elimination mechanism** with OH^- as a base: lone pair on O → H on the carbon next to C–X; the C–H bond pair → the C–C bond to make C=C; the C–X bond pair → X (WCH12 Jan 2022). See the figure above.
- Silver(I) oxide in water as a source of OH^-: $Ag_2O + H_2O \rightarrow 2Ag^+ + 2OH^-$; it's less corrosive than NaOH and can be recovered (WCH12 Jun 2023).
- Volumes, limiting reagent and % yield for a halogenoalkane preparation using density (WCH12 Jun 2019, Jun 2022).

COMMON MISTAKE

Writing "cyanide" or "hydroxide" without the metal: give **KCN** or **KOH** (WCH12 Oct 2020 report).

Missing "aqueous" for substitution, or "ethanol" / "under pressure" for the amine (WCH12 Jan 2024 report).

Suggesting NaOH to dehydrate an **alcohol**: that's halogenoalkane chemistry. Alcohols need conc. H_3PO_4 (10.17).

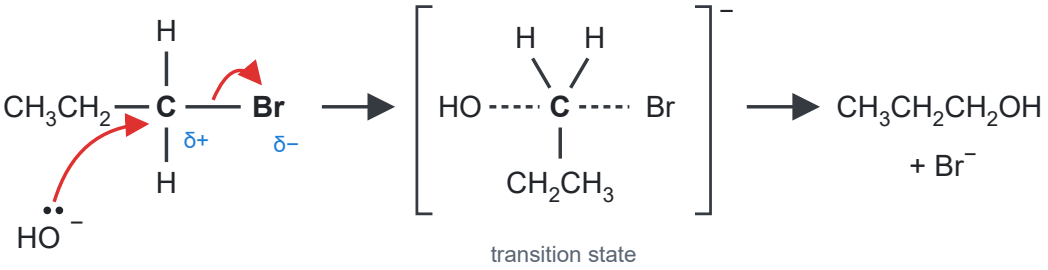
Drawing the nitrile as C–N or C=N, or naming it an amine or a "cyanide".

SPEC 10.9 Nucleophilic substitution mechanisms (OH^- and NH_3)

WHAT THIS SPEC POINT ASKS

Understand the mechanisms of nucleophilic substitution of primary halogenoalkanes with (i) aqueous KOH and (ii) ammonia. (S_N1 and S_N2 are named and tested in Unit 4.)

With hydroxide ions



transition state

The lone pair on oxygen attacks the δ+ carbon from the side opposite the bromine. As the new O–C bond forms, the C–Br bond breaks and both its electrons go with the bromide ion.

1. Draw the **δ+** on the carbon and the **δ−** on the bromine of the C–Br bond.
2. The **lone pair on the oxygen** of OH^- is attracted to the δ+ carbon. Draw a curly arrow **from that lone pair to the carbon**.
3. At the same time the C–Br bond breaks **heterolytically**. Draw a curly arrow **from the middle of the C–Br bond to the Br** (or just past it).
4. Products: **CH_3CH_2OH** (the O is now bonded to C) and **Br^-**.

This happens in a single step: the new C–O bond forms as the C–Br bond breaks. (In Unit 4 you'll call this **S_N2**.)

With ammonia

Step 1: substitution

1. δ+ on C, δ− on Br.
2. Curly arrow from the **lone pair on the N of NH_3** to the δ+ carbon.
3. Curly arrow from the **C–Br bond to Br**.
4. Products: the intermediate ion **$CH_3CH_2NH_3^+$** (N now has four bonds and a **positive charge**) and **Br^-**.

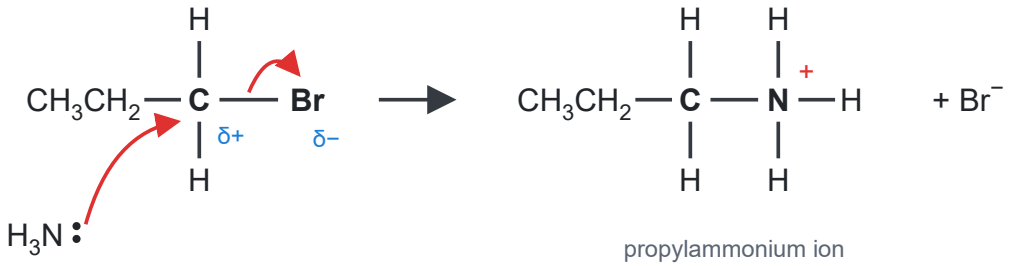
Step 2: loss of H^+

1. A **second NH_3** molecule uses its lone pair to attack one of the **H atoms on the N^+**: curly arrow from the lone pair on N to that H.
2. Curly arrow from the **N–H bond** to the **N^+**: the bond pair stays on the nitrogen, cancelling its charge.
3. Products: **ethylamine, $CH_3CH_2NH_2$**, and **NH_4^+** (which pairs with Br^- as NH_4Br).

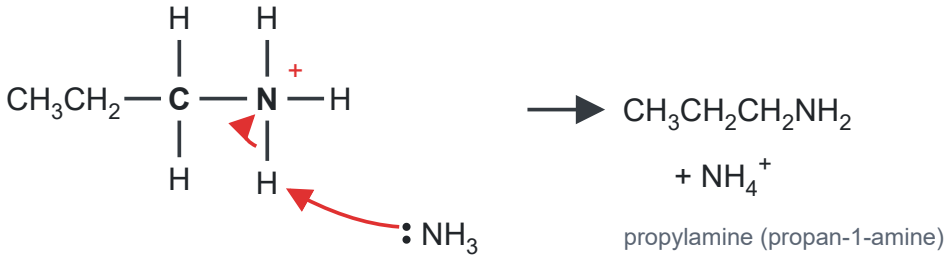
Overall: $CH_3CH_2Br + 2NH_3 \rightarrow CH_3CH_2NH_2 + NH_4Br$

Mechanism: 1-bromopropane with ammonia

1 The lone pair on the nitrogen of NH_3 attacks the δ+ carbon, and the C–Br bond breaks heterolytically. Nitrogen now has four bonds and no lone pair, so it carries a positive charge: the product of this step is the propylammonium ion, $CH_3CH_2CH_2NH_3^+$, plus Br^-.



2 A second ammonia molecule acts as a base. Its lone pair removes H^+ from the N–H bond of the ammonium ion, and the electrons of that N–H bond return to nitrogen as a lone pair.



The ammonium ion and the bromide ion end up together as ammonium bromide, NH_4Br, which is why two moles of ammonia appear in the overall equation.

EXAM TIP

Mechanism marks are given for **separate items**, so draw every one: δ+/δ−; each lone pair the arrow starts from (two dots on O or N); each arrow starting at a lone pair or bond; the charge on the intermediate; **every** product including Br^- and NH_4^+.

MARK-SCHEME WORDING

OH^- (WCH12 Jun 2019, Oct 2021):

- dipole on C–X and arrow from the **C–X bond to X** (or just beyond);
- arrow from the **lone pair on O of OH^-** to the δ+ C (not from the minus sign);

- both products. KCl is allowed if K^+ is shown. A covalent K–O bond in KOH loses the mark. A transition state is not needed.

NH_3 (WCH12 Jun 2025, six marking points for 3 marks):

- dipole on C–Br;
- arrow from the **lone pair on N** of NH_3 to the correct C;
- arrow from C–Br to Br;
- arrow from an **N–H bond of the intermediate to N**;
- arrow from the **lone pair on a second NH_3 to H** on the intermediate;
- **Br^- and NH_4^+** shown (or NH_4Br).

A missing lone pair, a charge on NH_3 and fish-hook arrows are each penalised once.

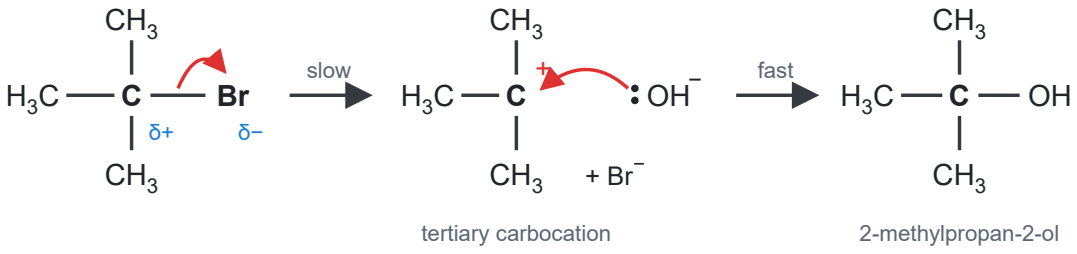
Often asked as: draw the mechanism (3–4 marks); find the omissions in a drawn mechanism ("lone pair on N", "δ+ on C").

⚠ Beyond the spec · not part of the Edexcel IAL specification

Examined in past papers even though the spec doesn't list it.

S_N1 for tertiary halogenoalkanes. WCH12 papers often use tertiary compounds and accept this mechanism (WCH12 Jun 2023, Oct 2021):

1. The C–Br bond in 2-bromo-2-methylpropane breaks **heterolytically** on its own: arrow from the C–Br bond to Br. This gives a **tertiary carbocation** $(CH_3)_3C^+$ (trigonal planar) and Br^-. This is the slow step.
2. The **lone pair on O of OH^-** (or H_2O) attacks the **C^+**: arrow from the lone pair to the positive carbon. Product: $(CH_3)_3COH$.



S_N1 for a tertiary halogenoalkane. The C–Br bond breaks first (slow) to leave a planar carbocation; the hydroxide ion then attacks it (fast).

Tertiary carbocations are the most stable because the three alkyl groups **push electron density** towards the C^+ (WCH12 Jan 2020). Primary halogenoalkanes react by the one-step route above.

The bond formed when NH_3 accepts H^+ in step 2 is a **dative covalent** bond: N's lone pair goes into the empty orbital of H^+ (WCH12 Jun 2021).

COMMON MISTAKE

Drawing ammonia with a **negative charge**, or the lone pair on an H instead of N (WCH12 Oct 2020, Jun 2022, Jan 2024 reports).

Missing the second NH_3 step, or leaving out Br^- and NH_4^+.

Arrows that show an H atom moving (from H to N) instead of the electron pair.

Arrows aimed at the C–C bond instead of the carbon atom.

SPEC 10.10 Comparing rates of hydrolysis from experimental data

WHAT THIS SPEC POINT ASKS

Use observations and data to compare the relative rates of hydrolysis of (i) primary, secondary and tertiary isomers of a halogenoalkane, and (ii) primary chloro-, bromo- and iodoalkanes, using aqueous silver nitrate in ethanol.

How the comparison works

- The halogenoalkane is warmed with **aqueous silver nitrate dissolved in ethanol**. Water slowly hydrolyses it, releasing X^- ions:
  - $R–X + H_2O \rightarrow R–OH + H^+ + X^-$
  - $Ag^+(aq) + X^-(aq) \rightarrow AgX(s)$
- The **time for a precipitate to first appear** measures the rate: **shorter time → faster hydrolysis**. Rate ∝ 1/time (9.3).
- The precipitate colour tells you which halogen was present.

Halogenoalkane	Precipitate	Colour	Speed
1-chlorobutane	AgCl	**white**	slowest (may take a long time)
1-bromobutane	AgBr	**cream**	faster
1-iodobutane	AgI	**yellow**	fastest

Isomer of C_4H_9Br	Class	Speed
1-bromobutane	primary	slowest
2-bromobutane	secondary	in between
2-bromo-2-methylpropane	tertiary	fastest (precipitate almost at once)

- To make the comparison fair, keep the **same volume (or number of drops)** of each halogenoalkane, the **same volume and concentration** of $AgNO_3$, the **same volume of ethanol** and the **same temperature** (water bath).
- **Fluoroalkanes** give no precipitate in a reasonable time: the C–F bond is too strong.

MARK-SCHEME WORDING

- What to measure: "**time taken for the (first appearance of the) precipitate** (of silver halide) to form" (WCH12 Oct 2021).
- 6-mark (WCH12 Oct 2020) on 1-bromobutane, 2-bromobutane and 2-bromo-2-methylbutane: nucleophilic substitution; silver halide precipitate; time depends on C–halogen **bond strength**; bond enthalpy decreases down Group 7 (larger atoms); rate **1° < 2° < 3°**; classify each compound. Wrong precipitate colours, or reasons based on **electronegativity** or **intermolecular forces**, lose a mark.
- Two opposing factors (WCH12 Jun 2022): 2-chloro-2-methylpropane "should react faster **because it is tertiary**"; 1-iodobutane "should react faster because the **C–I bond enthalpy is lower** than C–Cl"; so "it is **not possible to predict**" which wins.

Often asked as: describe how to compare rates; predict and explain the order; interpret times or a table of results; identify the halogen from the precipitate colour.

COMMON MISTAKE

"A stronger bond makes a faster reaction" (WCH12 Jan 2022 report): the **weaker** C–X bond breaks more easily.

Swapping the precipitate colours, or giving only formulae when colours were asked.

Describing the practical when the question asked you to compare and explain.

Calling the halogenoalkane itself a "carbocation".

SPEC 10.11 Core Practical 5: rates of hydrolysis of halogenoalkanes

WHAT THIS SPEC POINT ASKS

Investigate the rates of hydrolysis of some halogenoalkanes.

CORE PRACTICAL

Core practical 5: method in brief

Part 1: chloro v bromo v iodo

1. Label three test tubes. Put **1 cm³ of ethanol** in each.
2. Add **the same number of drops** (e.g. 3) of 1-chlorobutane, 1-bromobutane and 1-iodobutane, one to each tube.
3. Stand the tubes, and a tube of **0.05 mol dm⁻³ silver nitrate solution**, in a **water bath at about 50–60 °C** for a few minutes so everything reaches the same temperature.
4. Add **1 cm³ of the warm silver nitrate** to each tube, shake, and **start the clock**.
5. Record the **time for a precipitate to first appear** in each tube, and its colour.

Part 2: primary v secondary v tertiary

Repeat with 1-bromobutane, 2-bromobutane and 2-bromo-2-methylpropane (same halogen, so only the structure changes).

Variables

Independent	Dependent	Control
the halogen (part 1) or the class of halogenoalkane (part 2)	time for a precipitate to appear	volume of halogenoalkane, volume of ethanol, volume and concentration of $AgNO_3$, temperature of the water bath, size of test tubes

Processing results

WORKED EXAMPLE

Worked example: from times to relative rates

Times for a precipitate: 1-iodobutane 30 s, 1-bromobutane 240 s, 1-chlorobutane no precipitate after 1200 s.

Rate $\propto$ 1/t: iodo 1/30 = 0.033 s^{-1}; bromo 1/240 = 0.0042 s^{-1}; chloro < 1/1200 = 0.00083 s^{-1}.

Iodo is about 8 times faster than bromo. Order: **C–I > C–Br > C–Cl**, matching bond enthalpies 228 < 290 < 346 kJ mol^{-1}.

(These numbers are illustrative; your class results will differ, but the order shouldn't.)

Safety

- **Ethanol** and halogenoalkanes are **flammable**: heat with a water bath or electric heater, **no naked flames**.
- Halogenoalkanes are **harmful**; avoid skin contact and breathing the vapour.
- Silver nitrate is an **irritant** and **stains skin**; wear gloves and eye protection.

Errors and improvements

- Judging the **first appearance** of a precipitate is subjective, especially for the faint cream and yellow ones: view against a **black background**, or use a **colorimeter** / light sensor.
- Drops aren't all the same size: use a **graduated pipette** to add equal volumes (or equal **moles**, since the compounds have different densities and M_r).
- Temperature drops when the tubes are taken out: keep them **in the water bath** while timing.
- **Repeat** and take a mean.

EXAM TIP

The water (not OH^-) is the nucleophile here. That makes the reaction slow enough to compare by eye. If you used NaOH(aq) first, you'd have to **acidify with dilute nitric acid** before adding $AgNO_3$, otherwise brown silver oxide precipitates.

MARK-SCHEME WORDING

- "Time for the precipitate to form" (ALLOW "how fast"); "same temperature (water bath)"; "same volume/amount of halogenoalkane"; colours AgCl white, AgBr cream, AgI yellow.

Often asked as: how to measure the rate; identify control variables; predict the times for an unfamiliar halogenoalkane; explain the order (10.12, 10.13).

SPEC 10.12 Reactivity trend: primary, secondary and tertiary

WHAT THIS SPEC POINT ASKS

Know the trend in reactivity of primary, secondary and tertiary halogenoalkanes.

The trend

- For the same halogen, the rate of hydrolysis is **tertiary > secondary > primary**.
- Example: 2-bromo-2-methylpropane gives a cream precipitate with $AgNO_3$ in ethanol almost straight away; 1-bromobutane takes much longer.

Why (beyond the spec, but useful)

- Tertiary halogenoalkanes react by forming a **carbocation** first (S_N1, see 10.9). A **tertiary carbocation is the most stable**, because its three alkyl groups push electron density towards the C^+ and spread out the positive charge. So it forms most easily.
- Primary halogenoalkanes can't form a stable carbocation, so they react by the one-step route, where the nucleophile must reach the carbon directly.

MARK-SCHEME WORDING

- "**tertiary** halogenoalkanes react more quickly than primary" (WCH12 Jun 2024). ALLOW "tertiary carbocation is more stable than primary" and "methyl stabilisation of the positive charge" (WCH12 Jan 2022).
- DO NOT AWARD "tertiary halogenoalkanes react faster because they are **more stable**" (it's the **carbocation** that is more stable). IGNORE "more branched".

Often asked as: put isomers in order of rate; explain which of two compounds reacts faster (often with a second factor, the halogen, as well).

COMMON MISTAKE

Saying the tertiary **halogenoalkane** is more stable. It's the tertiary **carbocation** that is stabilised.

For a "state" question, writing a long carbocation explanation: just give the order (WCH12 Oct 2024 report).

SPEC 10.13 Reactivity trend: chloro-, bromo- and iodoalkanes (bond enthalpy)

WHAT THIS SPEC POINT ASKS

Explain, using bond enthalpy, the trend in reactivity of chloro-, bromo- and iodoalkanes.

Bond enthalpy decides

Bond	Mean bond enthalpy / kJ mol^{-1}	Polarity	Rate of hydrolysis
C–F	467	most polar	no reaction
C–Cl	346		slow
C–Br	290		faster
C–I	228	least polar	fastest

- Down Group 7 the halogen atom gets **larger**, so the C–X bond is **longer** and the shared pair is further from the halogen nucleus. The bond gets **weaker** (lower bond enthalpy).
- A weaker bond needs **less energy to break**, so the activation energy is lower and more collisions succeed. **Iodoalkanes react fastest.**
- **Polarity** would predict the opposite order: C–F is the most polar, so its carbon is the most δ+ and should attract nucleophiles best. The observed order shows that **bond enthalpy matters more than polarity**.

MARK-SCHEME WORDING

- "the **C–I bond is weaker** than C–Cl" / "C–I has a **lower bond enthalpy**", so iodoalkanes react faster (WCH12 Oct 2024, Jun 2024, Jan 2022).
- "C–Cl has a higher bond enthalpy (than C–Br)" so 1-chlorobutane is slower (WCH12 Jan 2022). Mark points may depend on the correct order.

Often asked as: explain why an iodoalkane reacts faster than a chloroalkane; combine with 10.12 for two compounds that differ in both class and halogen.

COMMON MISTAKE

Explaining with **electronegativity** or polarity, or with "reactivity of the halogen" or "size of the element" without linking to bond strength (WCH12 Jan 2025 report).

Vague "the iodine bond is weaker": say "the **C–I** bond".

SPEC 10.14 Core Practical 6: chlorination of 2-methylpropan-2-ol

WHAT THIS SPEC POINT ASKS

Prepare 2-chloro-2-methylpropane from 2-methylpropan-2-ol and concentrated hydrochloric acid.

The reaction

$(CH_3)_3COH + HCl \rightarrow (CH_3)_3CCl + H_2O$

- 2-methylpropan-2-ol is a **tertiary** alcohol. Tertiary alcohols react with **concentrated HCl at room temperature**, because they form a stable tertiary carbocation. Primary and secondary alcohols barely react this way (use PCl_5 for those: 10.17).
- 2-chloro-2-methylpropane boils at about **51 °C** and has density about 0.84 g cm^{-3}, so it forms the **upper** layer above the aqueous acid.

CORE PRACTICAL

Core practical 6: method in brief

1. In a **fume cupboard**, put about 10 cm^3 of 2-methylpropan-2-ol and about 35 cm^3 of **concentrated hydrochloric acid** into a **separating funnel**.
2. Stopper, **shake** for several minutes (about 20 minutes in total), **inverting and opening the tap often** to release the pressure.
3. Leave to stand until two layers separate. Run off and discard the **lower (aqueous) layer**.
4. Add **sodium hydrogencarbonate solution** to the organic layer to **neutralise leftover acid**. Shake gently, venting often: CO_2 is produced. Discard the aqueous layer. Wash once with water.
5. Run the organic layer into a dry conical flask and add **anhydrous sodium sulfate** (or $MgSO_4$ / $CaCl_2$). Swirl until the liquid is **clear**.
6. Decant or filter into a clean flask and **distil**, collecting the fraction that boils at about **49–53 °C**.

Why each step

Step	Purpose
Separating funnel and shaking	the two liquids don't mix; shaking brings them into contact
Releasing pressure	HCl gas and (later) CO_2 build up pressure
$NaHCO_3$ wash	removes acid: $NaHCO_3 + HCl \rightarrow NaCl + H_2O + CO_2$
Anhydrous salt	removes **water** left in the organic layer
Distillation	separates the product from unreacted alcohol (b.p. 82 °C) and other impurities

WORKED EXAMPLE

Worked example: percentage yield

10.0 cm^3 of 2-methylpropan-2-ol (density 0.786 g cm^{-3}, M_r 74.0) gave 6.50 g of 2-chloro-2-methylpropane (M_r 92.5).

Mass of alcohol = 10.0 × 0.786 = 7.86 g

Moles of alcohol = 7.86 ÷ 74.0 = 0.1062 mol (1 : 1 ratio, HCl in excess)

Theoretical mass of product = 0.1062 × 92.5 = 9.83 g

% yield = 6.50 ÷ 9.83 × 100 = **66.2%**

Yield is lost when the layers are separated, in the washing, in the drying agent and in the distillation flask.

MARK-SCHEME WORDING

- Drying agent: "(anhydrous) **sodium sulfate** / magnesium sulfate / calcium chloride / calcium sulfate / calcium oxide" (ALLOW silica gel). DO NOT AWARD concentrated sulfuric acid, phosphoric acid, $CuSO_4$ or $CaCO_3$ (WCH12 Jun 2019).

Often asked as: name a drying agent; explain a step ($NaHCO_3$ wash, venting, distillation range); calculate the yield. No WCH12 paper (2019–2026) asked for the whole method.

COMMON MISTAKE

Offering **anhydrous copper(II) sulfate** as a drying agent: it's a test for water, not a drying agent (WCH12 Jun 2019 report).

Giving two drying agents where one is wrong.

Not venting the funnel after adding $NaHCO_3$.

SPEC 10.15 Naming and drawing alcohols

WHAT THIS SPEC POINT ASKS

Name alcohols and draw their structural, displayed and skeletal formulae.

Naming rules

- The functional group is **–OH** (hydroxyl). General formula of the alcohol series: $\mathbf{C_nH_{2n+1}OH}$.

- Name = stem for the longest chain containing the C–OH + **-an-** + **position number** + **-ol**: propan-1-ol, propan-2-ol.
- Number the chain to give the **–OH the lowest number** (it takes priority over alkyl and halogen prefixes).
- Two –OH groups: keep the "e": ethane-1,2-diol.
- Branches go in front as prefixes: 2-methylpropan-2-ol.

Name	Structural formula
methanol	CH_3OH
ethanol	CH_3CH_2OH
propan-1-ol	$CH_3CH_2CH_2OH$
propan-2-ol	$CH_3CH(OH)CH_3$
butan-2-ol	$CH_3CH(OH)CH_2CH_3$
2-methylpropan-1-ol	$(CH_3)_2CHCH_2OH$
2-methylpropan-2-ol	$(CH_3)_3COH$
4-methylhexan-2-ol	$CH_3CH(OH)CH_2CH(CH_3)CH_2CH_3$
ethane-1,2-diol	$HOCH_2CH_2OH$

- **Displayed formula**: show the **O–H bond** as well as every C–H; the O must join to C (C–O–H), not H to C.
- **Skeletal formula**: zig-zag carbon chain with **OH** written at the end of a line from the right carbon (write "OH", not just "O").

MARK-SCHEME WORDING

- "4-methylhexan-2-ol" (ALLOW 4-methyl-2-hexanol, 4-methylhexane-2-ol); DO NOT AWARD "4-methylhex-2-ol" (WCH12 Jun 2021).
- "2-methylpropan-1-ol", "butan-2-ol", "2-methylpropan-2-ol" (WCH12 Oct 2019). If a skeletal formula is asked for, displayed formulae aren't credited.
- A molecular formula like C_3H_7OH is not a structure and scores zero (WCH12 Jun 2023).

Often asked as: name or draw isomeric alcohols ($C_4H_{10}O$, $C_6H_{13}OH$); give the skeletal formula of a named alcohol.

COMMON MISTAKE

"3-methylpropan-1-ol": a methyl on the end carbon just lengthens the chain. It's **butan-1-ol** (WCH12 Oct 2019 report). The branched one is **2-methylpropan-1-ol**.

Leaving out the position number (butanol for butan-1-ol); drawing the same isomer twice (just rotated or flipped); drawing a cyclic alcohol when the formula doesn't allow it.

Joining the OH by its H in a displayed or skeletal formula.

SPEC 10.16 Primary, secondary and tertiary alcohols

WHAT THIS SPEC POINT ASKS

Distinguish primary, secondary and tertiary alcohols.

Classifying

Class	C–OH carbon bonded to	H atoms on the C–OH carbon	Example
Primary	one other C (or none, methanol)	two (three for methanol)	ethanol, propan-1-ol
Secondary	two other C	one	propan-2-ol, butan-2-ol
Tertiary	three other C	none	2-methylpropan-2-ol

The four alcohols with formula C_4H_9OH:

Name	Class
butan-1-ol	primary
2-methylpropan-1-ol	primary
butan-2-ol	secondary
2-methylpropan-2-ol	tertiary

- The class decides what oxidation gives (10.18): **primary → aldehyde → carboxylic acid; secondary → ketone; tertiary → no reaction**. The number of **H atoms on the C–OH carbon** is what matters, because oxidation removes one of them.

MARK-SCHEME WORDING

- Secondary: "the carbon (atom) attached to the OH group is attached to **two other carbon atoms / alkyl groups**", or "has **one hydrogen** atom" (WCH12 Oct 2025). DO NOT AWARD "hydroxide group" or any

mention of a carbocation.

- Jan 2026: define all three classes (by H atoms or by R groups) and give a correct example of each.

Often asked as: classify an alcohol; draw one example of each class for a given formula; explain why a tertiary alcohol isn't oxidised.

COMMON MISTAKE

"Primary means the OH is on the end carbon" (WCH12 Oct 2022 report). It's about how many **carbons** are attached to the C–OH carbon. 2-methylpropan-1-ol has a branched chain but is still primary.

Writing "hydroxide" for the –OH group: hydroxide is the OH^- ion.

SPEC 10.17 Reactions of alcohols: combustion, halogenation, dehydration

WHAT THIS SPEC POINT ASKS

Know the reactions of alcohols with (i) oxygen (combustion), (ii) halogenating agents: PCl_5 (also a test for –OH), 50% concentrated sulfuric acid with KBr, and red phosphorus with iodine, and (iii) concentrated phosphoric acid to make alkenes by elimination. Mechanisms not needed.

(i) Combustion

- Alcohols burn in plenty of oxygen to give **CO_2 and H_2O**:
  - $CH_3CH_2OH(l) + 3O_2(g) \rightarrow 2CO_2(g) + 3H_2O(l)$
  - $C_3H_7OH(l) + 4½O_2(g) \rightarrow 3CO_2(g) + 4H_2O(l)$
- Balance O last: remember the alcohol already contains one O.
- Ethanol is used as a (bio)fuel; it burns with a clean blue flame.

(ii) Halogenation: replacing –OH with a halogen

Halogenoalkane made	Reagent and conditions	Equation (ethanol, or as stated)
Chloroalkane	**PCl_5**, room temperature, dry	$CH_3CH_2OH + PCl_5 \rightarrow CH_3CH_2Cl + POCl_3 + HCl$
Bromoalkane	**KBr + 50% concentrated H_2SO_4**, heat (HBr made in situ)	$KBr + H_2SO_4 \rightarrow KHSO_4 + HBr$; $CH_3CH_2OH + HBr \rightarrow CH_3CH_2Br + H_2O$

Halogenoalkane made	Reagent and conditions	Equation (ethanol, or as stated)
Iodoalkane	**red phosphorus + iodine**, warm under reflux (PI_3 made in situ)	$2P + 3I_2 \rightarrow 2PI_3$; $3CH_3CH_2OH + PI_3 \rightarrow 3CH_3CH_2I + H_3PO_3$
Chloroalkane (**tertiary** only)	**concentrated HCl**, room temperature	$(CH_3)_3COH + HCl \rightarrow (CH_3)_3CCl + H_2O$ (Core Practical 6)

- **PCl_5 as a test for –OH**: add solid PCl_5 to the dry sample. **Steamy (misty) white fumes of HCl** are given off. They turn **damp blue litmus red** and form **white smoke** (NH_4Cl) with ammonia on a glass rod.
  - Water and carboxylic acids (which also contain –OH) give the same result, so the sample must be **dry**, and the test shows –OH, not specifically an alcohol.
- **Why 50% sulfuric acid?** Concentrated H_2SO_4 is an oxidising agent: it would oxidise some of the HBr to **bromine** (Topic 8). Diluting it to about 50% reduces this.

(iii) Dehydration (elimination) with concentrated phosphoric acid

$CH_3CH_2OH \rightarrow CH_2{=}CH_2 + H_2O$ (conc. H_3PO_4, heat; write the catalyst **above the arrow**)

- A molecule of **water** is eliminated from neighbouring carbons (the –OH and an H next to it), leaving a C=C.
- Concentrated H_3PO_4 is a **catalyst**: it's preferred to conc. H_2SO_4 because it causes fewer side reactions (H_2SO_4 is oxidising and chars the alcohol).
- The alkene is usually **distilled off** as it forms.
- If the C–OH carbon has two different neighbouring carbons with H atoms, you get a mixture: butan-2-ol → **but-1-ene + E-but-2-ene + Z-but-2-ene**.

MARK-SCHEME WORDING

- PCl_5 test: "**misty fumes** evolved (that turn damp blue litmus red / form white smoke with ammonia)". ALLOW steamy / white fumes; PCl_3 allowed. DO NOT AWARD "white smoke" unless ammonia is involved. IGNORE "gas / hydrogen given off" (WCH12 Jun 2019).
- Equation: "$C_5H_{11}OH + PCl_5 \rightarrow C_5H_{11}Cl$ + **$POCl_3$** + **HCl**" (ACCEPT PCl_3O) (WCH12 Jan 2020).
- Bromination: "**50% / concentrated sulfuric acid and potassium bromide**". ALLOW P + Br_2, PBr_3, PBr_5. DO NOT AWARD dilute sulfuric acid (WCH12 Jun 2021).
- Chlorination: PCl_5, or **concentrated** HCl, $SOCl_2$, PCl_3. IGNORE just "HCl" (WCH12 Jun 2021).
- Dehydration: "(concentrated) **phosphoric(V) acid** / H_3PO_4"; ACCEPT concentrated H_2SO_4 or aluminium oxide. DO NOT AWARD dilute sulfuric acid (WCH12 Jan 2020).

Often asked as: reagents for converting an alcohol to a halogenoalkane or alkene; equations; the test for –OH; 6-mark choice of the best test for an alcohol group.

⚠ Beyond the spec · not part of the Edexcel IAL specification

Examined in past papers even though the spec doesn't list it.

- Making a chloroalkane from an alcohol with PCl_5 is better than adding HCl to an alkene: it's **one step** and gives **no isomeric by-products** (WCH12 Jan 2020).
- Choosing a test for –OH (WCH12 Jun 2024): acidified dichromate fails for a **tertiary** alcohol; acidified $KMnO_4$ also reacts with **C=C**; PCl_5 works for all three classes.
- Other –OH tests: **sodium** gives effervescence (H_2); Lucas' reagent turns cloudy at once with tertiary alcohols (WCH12 Jun 2019).
- Bioethanol: renewable; fermentation must be **anaerobic** so the ethanol isn't oxidised; water in fuel ethanol causes corrosion and lowers energy per litre (WCH12 Oct 2021).

COMMON MISTAKE

"Add HCl" without "**concentrated**" (WCH12 Jan 2021 report).

Suggesting **NaOH** to dehydrate an alcohol (WCH12 Jan 2020 report).

Drawing the same alkene twice (rotated) as two isomers; claiming E/Z isomers for an alkene that can't have them.

SPEC 10.18 Oxidation of alcohols with acidified potassium dichromate(VI)

WHAT THIS SPEC POINT ASKS

Know that acidified potassium dichromate(VI) oxidises (i) primary alcohols to aldehydes (positive with Benedict's or Fehling's) if distilled as they form, (ii) primary alcohols to carboxylic acids (positive with Na_2CO_3 or $NaHCO_3$) if heated under reflux, and (iii) secondary alcohols to ketones. [O] may be used in equations.

The oxidising agent

- **Potassium dichromate(VI), $K_2Cr_2O_7$, in dilute sulfuric acid**. Name the acid: "acidified" alone isn't enough.
- Colour change: **orange** ($Cr_2O_7^{2-}$, Cr +6) → **green** (Cr^{3+}, Cr +3). The dichromate is **reduced** while the alcohol is oxidised.
- Half-equation: $Cr_2O_7^{2-} + 14H^+ + 6e^- \rightarrow 2Cr^{3+} + 7H_2O$
- In organic equations write the oxidising agent as **[O]**.

What each class gives

Alcohol	Product	Conditions	Equation (example)
Primary	**aldehyde**	**distil off as it forms**; alcohol in excess	$CH_3CH_2CH_2OH + [O] \rightarrow CH_3CH_2CHO + H_2O$
Primary	**carboxylic acid**	**heat under reflux**; **excess** oxidising agent	$CH_3CH_2CH_2OH + 2[O] \rightarrow CH_3CH_2COOH + H_2O$
Secondary	**ketone**	heat under reflux	$CH_3CH(OH)CH_3 + [O] \rightarrow CH_3COCH_3 + H_2O$
Tertiary	**no reaction**	stays orange	–

- **Aldehyde v acid: why the conditions matter.** The aldehyde is oxidised further to the acid if it stays in the flask. An aldehyde has a much **lower boiling point** than the alcohol (aldehyde molecules can't hydrogen-bond to each other), so it **distils out as soon as it forms** and escapes further oxidation. Under **reflux**, the aldehyde vapour condenses and runs back, so it is oxidised all the way to the acid.
- **Tertiary alcohols** have **no H atom on the C–OH carbon**, so they can't be oxidised this way: the mixture stays orange.
- **Ketones** aren't oxidised further, for the same reason.

Testing the products

Product	Test	Positive result
Aldehyde	warm with **Benedict's or Fehling's solution**	**blue** solution → **red(-brown) precipitate** (Cu_2O)
Ketone	Benedict's / Fehling's	stays blue (no reaction)
Carboxylic acid	add **sodium carbonate** or **sodium hydrogencarbonate**	**effervescence**; the gas (CO_2) turns **limewater milky**

- Acid with hydrogencarbonate: $CH_3CH_2COOH + NaHCO_3 \rightarrow CH_3CH_2COONa + H_2O + CO_2$

MARK-SCHEME WORDING

- Reagents for the acid: "sodium / potassium **dichromate(VI)" and "(dilute) sulfuric acid" and** "(heat under) **reflux**". DO NOT AWARD potassium manganate(VII) or hydrochloric acid (WCH12 Oct 2020, Jun 2025).
- For the aldehyde: "(heat with) potassium dichromate(VI) and sulfuric acid", "**distil**", "so the aldehyde is **not oxidised further** / does not form a carboxylic acid". DO NOT AWARD reflux (WCH12 Jan 2026).
- Equations: "$CH_3OH + 2[O] \rightarrow HCOOH + H_2O$"; "$CH_3CH_2CH_2OH + 2[O] \rightarrow CH_3CH_2COOH + H_2O$"; going via the aldehyde in two equations is allowed.

- Colour change: "**orange to green**" (ALLOW orange to blue); DO NOT AWARD the reverse.
- Tertiary alcohols "cannot be oxidised"; a primary alcohol in the body is oxidised to a toxic aldehyde (WCH12 Jan 2020).

Often asked as: reagents and conditions for an aldehyde or acid; equations with [O]; colour change; tests for the products; why a tertiary alcohol doesn't react; 6-mark on making propanoic acid.

⚠ Beyond the spec · not part of the Edexcel IAL specification

Examined in past papers even though the spec doesn't list it.

- 6-mark (WCH12 Jun 2019): higher concentration or temperature → faster rate (more frequent collisions, more particles with $E \geq E_a$); **excess** oxidising agent and **reflux** ensure **complete** oxidation to the acid.
- IR can follow the oxidation: the **O–H** peak (3750–3200 cm^{-1}) shrinks and a **C=O** peak (1740–1700 cm^{-1}) appears (WCH12 Oct 2025).
- The orange/green chromium colours mask an indicator, so the acid made can't be titrated directly in the reaction mixture (WCH12 Jun 2019).

COMMON MISTAKE

"Acidified potassium dichromate" with **no acid named** (WCH12 Jan 2025 report).

Using **reflux** to make an aldehyde (WCH12 Jan 2025 report).

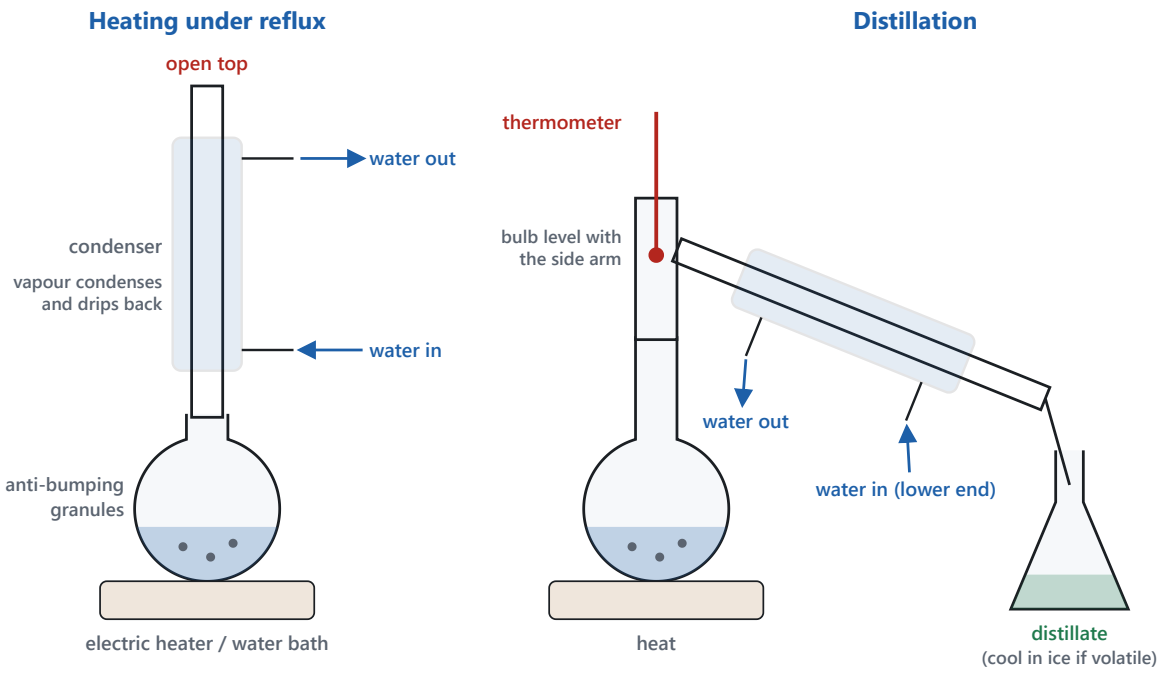
Colour change backwards, or a wrong oxidation number for Cr.

Forgetting that tertiary alcohols **don't react** (WCH12 Oct 2022 report: over half scored nothing).

SPEC 10.19 Techniques for preparing and purifying a liquid organic product

WHAT THIS SPEC POINT ASKS

Understand (i) heating under reflux, (ii) solvent extraction with a separating funnel, (iii) distillation, (iv) drying with an anhydrous salt, (v) measuring a boiling temperature.



Left: reflux (vertical condenser, open at the top). Right: distillation (thermometer bulb level with the side arm; water enters the condenser at the lower end).

(i) Heating under reflux

- **What**: heat the mixture in a flask with a **vertical condenser** on top.
- **Why**: organic reactions are slow and need heating for a long time. The volatile reactants and products **evaporate, condense and drip back** into the flask, so nothing is lost and the reaction can go to completion.
- Details:
  - add **anti-bumping granules** for smooth boiling;
  - water goes **in at the bottom** of the condenser and out at the top, so the jacket stays full;
  - the top is **open**: never heat a sealed system (pressure would build up and it could explode);
  - heat with a **water bath or electric heater**, not a Bunsen, because organic vapours are flammable.

(ii) Extraction with a separating funnel

- Used to separate two liquids that **don't mix** (immiscible), e.g. an organic layer from an aqueous layer, or to **wash** an organic product.
- Shake with the stopper on, **inverting and opening the tap** regularly to release pressure (especially with $NaHCO_3$, which makes CO_2).
- Let the layers settle. The **denser** liquid is the **lower** layer. Run it off through the tap.
- Not sure which layer is which? Add a few drops of **water**: the layer it joins is the aqueous one.
- **Washing**: shake with **sodium hydrogencarbonate** (or sodium carbonate) solution to remove acid, then with water.

(iii) Distillation

- Separates a liquid from **non-volatile** impurities or from liquids with **clearly different boiling points**.
- The **thermometer bulb** is level with the **side arm**, so it reads the temperature of the vapour that is actually distilling.
- **Sloping condenser**, water in at the **lower** end.
- **Collect the fraction** that boils within about **2 °C** of the product's boiling point; liquid coming over below or above that is a different substance.
- **Fractional distillation** (with a fractionating column) is used when boiling points are close.

(iv) Drying with an anhydrous salt

- Organic layers from a separating funnel contain a little water. Add an anhydrous salt: **anhydrous sodium sulfate, magnesium sulfate or calcium chloride**.
- Swirl. If the salt **clumps together**, add more; the liquid is dry when it turns from **cloudy to clear** and fresh salt moves freely.
- **Decant or filter** the dried liquid off the solid.

(v) Boiling temperature determination

- A **pure** liquid boils at a **fixed, sharp** temperature. Compare it with a data-book value to identify the product and check purity.
- **Impurities** change the boiling temperature and make it boil over a **range**.
- Method: distil the product (or heat a small sample in a tube in an oil bath) and record the steady thermometer reading while the liquid boils.

The usual purification order

reaction (reflux) → separate (separating funnel) → wash ($NaHCO_3$ / water) → dry (anhydrous salt) → distil (collect at the boiling point) → check purity (boiling temperature)

MARK-SCHEME WORDING

- Name the apparatus: "(reflux / Liebig) **condenser**". An answer that suggests a **sealed** system is not awarded (WCH12 Oct 2025).
- Why distillation fails to separate: "the boiling temperatures are **(too) similar / close**" (WCH12 Jun 2025).
- Why not reflux for the ammonia reaction: "ammonia (gas) would **escape** (from the condenser)" (WCH12 Jun 2022).
- Purifying bioethanol: "(fractional) distillation" (WCH12 Oct 2021).

Often asked as: reflux v distillation; name a piece of apparatus; why the system isn't sealed; choose a drying agent; why a separation won't work.

COMMON MISTAKE

Sealing the top of a reflux condenser.

Water entering the condenser at the top (the jacket never fills).

Thermometer bulb dipping into the liquid during distillation.

Copper(II) sulfate as a drying agent.

SPEC 10.20 Core Practical 7: oxidation of propan-1-ol to propanal and propanoic acid

WHAT THIS SPEC POINT ASKS

Oxidise propan-1-ol to make propanal (by distillation) and propanoic acid (by heating under reflux).

Boiling points decide the method

Compound	b.p. / °C	Why
propanal, CH_3CH_2CHO	49	no hydrogen bonding between its molecules
propan-1-ol, $CH_3CH_2CH_2OH$	97	hydrogen bonds
propanoic acid, CH_3CH_2COOH	141	hydrogen bonds (and a larger molecule)

CORE PRACTICAL

Core practical 7: method in brief

Propanal (distil as it forms)

1. Make the oxidising mixture: potassium dichromate(VI) dissolved in **dilute sulfuric acid**, with anti-bumping granules, in a flask set up for **distillation**.
2. Warm the mixture gently. Add **propan-1-ol** (in excess, so there isn't enough oxidant for a second oxidation) **dropwise** from a tap funnel.
3. Keep the thermometer reading **below about 60 °C** and collect the **distillate** in a receiver **cooled in ice** (propanal is very volatile).
4. Test the distillate with **Benedict's or Fehling's** solution: blue → **red precipitate** on warming shows an aldehyde.

Propanoic acid (heat under reflux)

1. Put propan-1-ol with an **excess** of acidified potassium dichromate(VI) and anti-bumping granules in a flask with a **reflux** condenser.
2. Heat under reflux for about 15–20 minutes. The colour turns from **orange to green**.
3. Rearrange for **distillation** and distil off the propanoic acid (with some water).
4. Test the distillate: add **sodium hydrogencarbonate** (or sodium carbonate): **effervescence**, gas turns limewater milky. Universal indicator shows it's weakly acidic.

Equations

$CH_3CH_2CH_2OH + [O] \rightarrow CH_3CH_2CHO + H_2O$
$CH_3CH_2CH_2OH + 2[O] \rightarrow CH_3CH_2COOH + H_2O$

Safety

- Potassium dichromate(VI) is **toxic, carcinogenic** and an oxidiser: wear gloves and eye protection.
- Propan-1-ol and propanal are **flammable**: use a water bath or electric heater.
- Concentrated reagents and hot liquids: anti-bumping granules, never heat a sealed apparatus.

MARK-SCHEME WORDING

- Propanal: "**distil**" so it is "not oxidised further"; reflux is not credited. Propanoic acid: "(heat under) **reflux**" with **excess** oxidant (WCH12 Jun 2021, Jan 2026).
- Colour change "orange to green"; condenser named; IR shows O–H (alcohol) disappearing and C=O appearing (WCH12 Oct 2025).
- Propanoic acid as a food preservative: below the limit "the food would go mouldy (it wouldn't preserve)", above it "the food tastes bad / becomes toxic" (WCH12 Jun 2019).

Often asked as: distil v reflux to get each product; equations; tests for each product; a Section C context on propanoic acid with a titration.

COMMON MISTAKE

Using reflux for propanal, or distillation straight away for propanoic acid.

"Acidified dichromate" without naming sulfuric acid.

SPEC 10.21 Mass spectra of organic compounds

WHAT THIS SPEC POINT ASKS

Use the m/z of the molecular ion and the fragmentation pattern in a mass spectrum to suggest possible structures of simple organic compounds.

Reading a spectrum

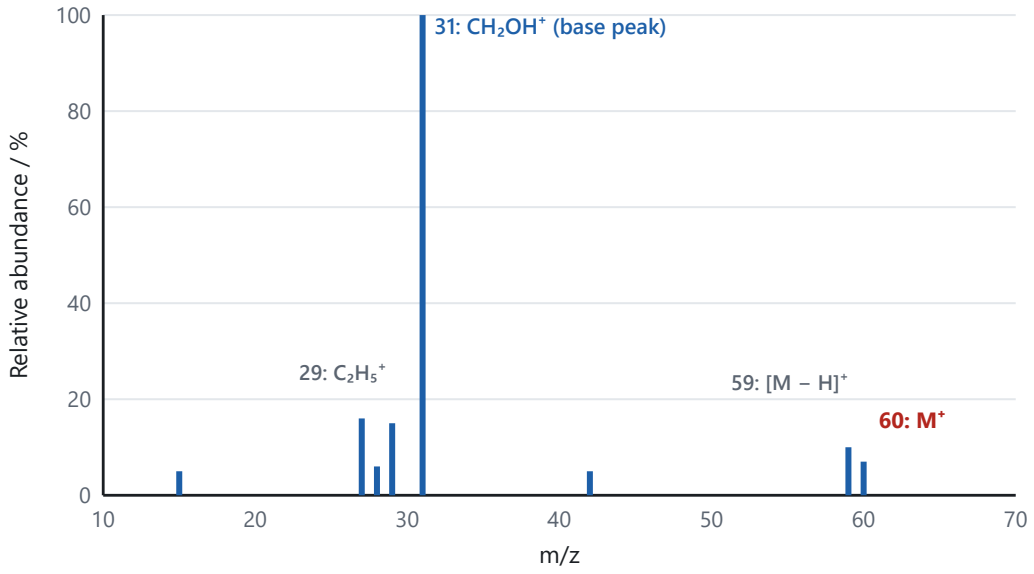
- Molecules are ionised by losing an electron: M → **M^+** + e^-. M^+ is the **molecular ion**.
- The **molecular ion peak** is the peak with the **highest m/z** (ignore the tiny M+1 peak from ^{13}C). Its m/z = **relative molecular mass, M_r**.
- Molecular ions are unstable and **fragment**: $M^+ \rightarrow X^+ + Y\bullet$. Only the **positive ion** X^+ is detected; the neutral radical Y• isn't.
- The **base peak** is the tallest peak (set to 100%): the most abundant (most stable) ion.
- The fragments show which groups are in the molecule. M_r alone only gives the molecular formula, and **isomers** have the same M_r: the fragments tell them apart.

Common fragment ions and losses

m/z	Fragment ion	Mass lost from M^+	Group lost
15	CH_3^+	15	CH_3
29	$C_2H_5^+$ or CHO^+	17	OH
31	CH_2OH^+	18	H_2O (alcohols)
43	$C_3H_7^+$ or CH_3CO^+	29	C_2H_5 or CHO
45	$COOH^+$ or CH_3CHOH^+	35 / 37	Cl
57	$C_4H_9^+$	79 / 81	Br

Halogen isotope patterns

- Chlorine is about **75% ^{35}Cl and 25% ^{37}Cl**, so a compound with **one Cl** shows two molecular-ion peaks **2 apart**, heights **3 : 1** (e.g. chloroethane: 64 and 66).
- Bromine is about **50% ^{79}Br and 50% ^{81}Br**, so **one Br** gives two peaks **2 apart** of **equal height**, **1 : 1** (e.g. bromomethane: 94 and 96).
- **Two Cl atoms** give M, M+2 and M+4 in the ratio **9 : 6 : 1**.



Simplified mass spectrum of propan-1-ol, $CH_3CH_2CH_2OH$ (M_r 60). The base peak at 31 is CH_2OH^+, which shows the –OH is on the end carbon.

WORKED EXAMPLE

Worked example: propan-1-ol or propan-2-ol?

Both have M_r = 60 (C_3H_8O), so the molecular ion at m/z 60 can't tell them apart.

Propan-1-ol, CH_3CH_2–CH_2OH: breaking the C–C bond next to the C–OH gives **CH_2OH^+, m/z 31** (base peak). Also $C_2H_5^+$ at 29.

Propan-2-ol, CH_3–$CH(OH)$–CH_3: losing a CH_3 gives **CH_3CHOH^+, m/z 45** (base peak). There is no big peak at 31.

So a large peak at 31 → propan-1-ol; a large peak at 45 → propan-2-ol.

WORKED EXAMPLE

Worked example: propanal or propanone?

Both C_3H_6O, M_r = 58.

Propanone, CH_3COCH_3: strong peak at **43** (CH_3CO^+, loss of CH_3) and at 15 (CH_3^+).

Propanal, CH_3CH_2CHO: strong peaks at **29** ($C_2H_5^+$ and CHO^+) and 28; 57 (loss of H).

EXAM TIP

When you identify a fragment, write its **formula with a + charge** ("CH_2OH^+", not "CH_2OH" and not just "m/z 31"). Missing charges are penalised.

MARK-SCHEME WORDING

- Fragments must be written as **positive ions** with a structure or formula, e.g. "m/z = 53 is $(CH_2{=}C{-}CH{=}CH_2)^+$, loss of CH_3" (WCH12 Jun 2024). Missing charges penalised once.
- Isotopes: "chlorine isotopes (35 and 37) are in the ratio **3 : 1**", so "two chlorine atoms give the ratio **9 : 6 : 1**" (WCH12 Jun 2021). DO NOT AWARD ^{36}Cl.

Often asked as: identify the species causing given peaks; use M^+ to choose between candidate compounds; explain a pattern of peaks for a compound containing Cl or Br; 6-mark combined MS + IR identification (WCH12 Jan 2025).

⚠ Beyond the spec · not part of the Edexcel IAL specification

Examined in past papers even though the spec doesn't list it.

- Find M_r first from gas data ($pV = nRT$, convert kPa to Pa and cm^3 to m^3) or from % composition, then match to M^+ (WCH12 Oct 2024, Jun 2025).
- Peaks for $COCl_2$ at 98, 100, 102 in the ratio 9 : 6 : 1, and the fragment $CO^{35}Cl^+$ (WCH12 Jun 2021). (That mark scheme writes the abundances the wrong way round: chlorine is 75% ^{35}**Cl**.)

COMMON MISTAKE

Taking the tallest peak (base peak) as the molecular ion.

Calculating the M_r of the candidate compounds wrongly (WCH12 Jan 2025 report): check your arithmetic.

Fragments written without a positive charge.

SPEC 10.22 Infrared spectra

WHAT THIS SPEC POINT ASKS

Use IR spectra (or IR data) to identify functional groups and predict absorptions for: C–H (alkanes, alkenes, aldehydes), C=C, O–H (alcohols, carboxylic acids), C=O (aldehydes, ketones, carboxylic acids), C–X and N–H (amines).

How IR works

- Covalent bonds **vibrate** (stretch and bend). Each type of bond absorbs **infrared radiation** at its own frequency, depending on the **bond strength** and the **masses of the atoms**.
- An IR spectrum plots **% transmittance** (y) against **wavenumber / cm^{-1}** (x, running from 4000 on the left down to about 500). Absorptions show as **dips** (downward "peaks").
- Describe a peak by **position** (wavenumber), **width** (broad or sharp) and **intensity** (strong or weak).
- Below about 1500 cm^{-1} is the **fingerprint region**: a complicated pattern unique to each compound. Matching it to a database spectrum confirms an identity.

Key absorptions (check against your Data Booklet)

Bond	Where	Wavenumber / cm^{-1}	Look
O–H	alcohols	3750–3200 (hydrogen-bonded: about 3550–3230)	strong, **broad**
O–H	carboxylic acids	3300–2500	**very broad**, overlaps C–H
N–H	amines	3500–3300	medium, sharper than O–H
C–H	alkenes (=C–H)	3095–3010	medium
C–H	alkanes	2962–2853	strong (in almost every organic compound)
C–H	aldehydes (–CHO)	2900–2820 and 2775–2700	two weak peaks
C=O	aldehydes	1740–1720	strong, **sharp**
C=O	ketones	1720–1700	strong, sharp
C=O	carboxylic acids	1725–1700	strong, sharp
C=C	alkenes	1669–1645	variable
C–O	alcohols, acids, esters	1300–1000	strong
C–Cl	chloroalkanes	about 800–600	strong
C–Br / **C–I**	bromo- / iodoalkanes	about 600–500 / about 500	

Using it

- Look first for the big diagnostic peaks: **O–H (broad, above 3200)**, **C=O (sharp, about 1700)**, **N–H**, **C=C**.
- Combine them:

Peaks seen	Compound type
broad O–H 3750–3200, no C=O	**alcohol**
C=O about 1720, no O–H	**aldehyde or ketone** (aldehyde if C–H at 2900–2700)
very broad O–H 3300–2500 **and** C=O about 1710	**carboxylic acid**
C=C 1669–1645 and =C–H above 3000	**alkene**
N–H 3500–3300	**amine**
C–X below 800, nothing else special	**halogenoalkane**

WORKED EXAMPLE

Worked example: following an oxidation

Propan-1-ol is heated under reflux with acidified $K_2Cr_2O_7$. How does the IR spectrum show that propanoic acid has formed?

Before: broad **O–H (alcohol)** absorption at **3750–3200 cm^{-1}**; no C=O.

After: a strong **C=O** absorption appears at **1725–1700 cm^{-1}**, and the O–H absorption moves to a **very broad band at 3300–2500 cm^{-1}** (carboxylic acid O–H).

If propanal formed instead (distillation), there would be **C=O at 1740–1720 cm^{-1}** and **no O–H** at all.

WORKED EXAMPLE

Worked example: an elimination

Butan-2-ol is dehydrated to butenes. IR evidence: the **O–H** absorption at **3750–3200 cm^{-1}** disappears; a **C=C** absorption appears at **1669–1645 cm^{-1}** (and =C–H at 3095–3010 cm^{-1}).

MARK-SCHEME WORDING

- Give **the bond and its wavenumber range**: "Both wavenumber ranges without bonds scores (1)"; a single specific wavenumber inside a range is accepted, but you still need the bond (WCH12 Jun 2023).

- Product v reactant: "O–H absent / C=O present". "One mark for two correct bonds identified." C–H peaks and the fingerprint region are ignored in these questions (WCH12 Jun 2025).
- Elimination: "O–H present in the reactant / absent in the product" and "C=C in the product" (WCH12 Jun 2019).
- Ranges are those in the **Data Booklet**; any value within the booklet range is accepted.

Often asked as: give one IR difference between a reactant and product; assign labelled peaks; choose the compound that matches a spectrum (6-mark with MS, WCH12 Jan 2025).

⚠ Beyond the spec · not part of the Edexcel IAL specification

Examined in past papers even though the spec doesn't list it.

- The **fingerprint region** changes when a reaction happens, which can be used to show it has occurred (WCH12 Oct 2025, Oct 2020).
- The C=O in an **acyl chloride** ($COCl_2$) absorbs at a higher wavenumber (about 1795 cm^{-1}) than in ketones; mark schemes accept a wide range for unfamiliar C=O compounds (WCH12 Jun 2021).

COMMON MISTAKE

Quoting a bond with no wavenumber, or a wavenumber with no bond.

Not using the Data Booklet ranges (WCH12 Jan 2025 report).

Confusing alcohol O–H (3750–3200) with carboxylic acid O–H (3300–2500, much broader).

Using C–H absorptions to tell compounds apart: nearly every organic compound has them.

SPEC 10.23 Core Practical 8: analysis of some inorganic and organic unknowns

WHAT THIS SPEC POINT ASKS

Use chemical tests to identify unknown inorganic and organic substances.

Inorganic tests

Testing for	Method	Positive result
Group 1 / 2 cations	**flame test**: clean nichrome wire dipped in conc. HCl, then the solid, into a hot (blue) flame	Li^+ red; Na^+ yellow; K^+ lilac; Ca^{2+} orange-red; Sr^{2+} red (crimson); Ba^{2+} pale green
NH_4^+	warm with **NaOH(aq)**	gas (NH_3) turns **damp red litmus blue**
CO_3^{2-} / HCO_3^-	add **dilute acid**	effervescence; CO_2 turns **limewater milky**
SO_4^{2-}	add **dilute HCl**, then **$BaCl_2$(aq)**	**white precipitate** ($BaSO_4$)
Cl^-, Br^-, I^-	add **dilute HNO_3**, then **$AgNO_3$(aq)**; then **NH_3(aq)**	AgCl **white**, dissolves in dilute NH_3; AgBr **cream**, dissolves in conc. NH_3; AgI **yellow**, insoluble in conc. NH_3

- The HCl (for sulfate) and HNO_3 (for halides) remove **carbonate** ions first, which would also give a precipitate.
- Use HNO_3, not HCl, before $AgNO_3$: HCl would add Cl^- ions.

Organic tests

Testing for	Reagent	Positive result
C=C (alkene)	**bromine water** (shake)	**orange → colourless**
–OH (alcohol, acid, water)	**PCl_5** (dry sample)	**steamy fumes** of HCl; white smoke with NH_3
1° or 2° alcohol, or aldehyde	warm with **acidified $K_2Cr_2O_7$**	**orange → green** (tertiary alcohols and ketones: stays orange)
Aldehyde	warm with **Benedict's or Fehling's**	**blue → red precipitate** (ketones: no change)
Carboxylic acid	**$NaHCO_3$** or Na_2CO_3	**effervescence**; CO_2 turns limewater milky
Halogenoalkane	warm with **NaOH(aq)**, add **dilute HNO_3**, then **$AgNO_3$(aq)**	precipitate: white (Cl), cream (Br), yellow (I)

CORE PRACTICAL

Core practical 8: approach

1. Record the **appearance** of each unknown (state, colour, smell only by wafting).
2. Plan a **sequence** of tests that narrows the options: start with tests that split the groups (e.g. flame test, then anion tests; for organics: bromine water, then PCl_5, then dichromate, then Benedict's/Fehling's or $NaHCO_3$).
3. Use **small amounts** (a few drops or a spatula tip) in test tubes; add reagents dropwise.
4. Record **observations** (what you see), not just conclusions. "Effervescence" not "CO_2 produced".
5. Combine results to identify each unknown. Use IR or mass spectra if they are given.

MARK-SCHEME WORDING

- Test for –OH (6-mark, WCH12 Jun 2024): **PCl_5** gives misty fumes with all alcohols; acidified dichromate fails for a tertiary alcohol; acidified $KMnO_4$ also reacts with C=C.
- Acid test: "effervescence" with carbonate (WCH12 Oct 2020). Aldehyde test: Benedict's or Fehling's.
- Halide: $AgNO_3$ with dilute nitric acid, then ammonia solution (WCH12 Jan 2021).
- Describe **observations**: "a white precipitate", "effervescence", "orange to green", not just the product's name.

Often asked as: choose a test to tell two compounds apart and give the result for each; 6-mark test choice; combine tests with spectra.

COMMON MISTAKE

Writing the name of a product ("silver chloride forms") when an **observation** is asked.

Giving the result for only **one** of the two compounds in a "distinguish" question: say what happens with **each**.

Testing a halogenoalkane directly with $AgNO_3(aq)$ with no hydrolysis step, or skipping the HNO_3 after NaOH.

Quick checks

QUICK CHECK

What is the name of $CH_3CHClCBr(CH_3)CH_3$?

- 3-chloro-2-bromo-2-methylbutane
- 2-bromo-3-chloro-2-methylbutane
- 2-methyl-2-bromo-3-chlorobutane
- 3-bromo-2-chloro-3-methylbutane

The longest chain has four carbons (butane). Numbering from the end nearest the Br gives the substituents positions 2, 2 and 3 rather than 2, 3 and 3. The prefixes then go in alphabetical order: bromo, chloro, methyl. Putting chloro first ignores the alphabetical rule.

QUICK CHECK

1-Chlorobutane, 1-bromobutane and 1-iodobutane are warmed with aqueous silver nitrate in ethanol. Which statement explains the order in which the precipitates appear?

- The C–Cl bond is the most polar, so 1-chlorobutane reacts first.
- Iodide ions are the best nucleophiles, so the iodo compound reacts first.
- Silver ions attack the C–I bond fastest because iodine is the largest atom.
- The C–I bond has the lowest bond enthalpy, so 1-iodobutane is hydrolysed fastest.

Water is the nucleophile, and breaking the C–X bond controls the rate. C–I (about 228 kJ mol^{-1}) is much weaker than C–Cl (346 kJ mol^{-1}), so the yellow AgI appears first. Polarity predicts the wrong order: C–Cl is more polar than C–I, yet slower.

QUICK CHECK

The rate equation for the hydrolysis of a halogenoalkane by hydroxide ions is rate = k[RBr]. Which halogenoalkane is it most likely to be?

- 1-bromobutane
- bromomethane
- 2-bromo-2-methylpropane
- 1-bromo-2-methylpropane

A rate that does not depend on [OH^-] means hydroxide is not in the slow step: the S_N1 route, where the C–Br bond breaks first to give a carbocation. Only the tertiary compound forms a stable enough carbocation. 1-Bromo-2-methylpropane is branched, but the carbon carrying the Br has only one other carbon on it, so it is primary.

QUICK CHECK

2-Bromopropane is heated under reflux with potassium hydroxide dissolved in ethanol. What is the main organic product and what is the role of the hydroxide ion?

- propan-2-ol; nucleophile
- propene; base
- propene; nucleophile
- propan-2-ol; base

In ethanol the hydroxide ion removes an H^+ from the carbon next to the C–Br carbon, so it acts as a base and an alkene forms by elimination. In water it would act as a nucleophile and give the alcohol.

QUICK CHECK

Which alcohol does *not* change the colour of acidified potassium dichromate(VI) when heated with it?

- butan-1-ol
- butan-2-ol
- 2-methylpropan-1-ol
- 2-methylpropan-2-ol

2-Methylpropan-2-ol is tertiary: the carbon carrying the –OH has no hydrogen, so no C=O can form and the dichromate stays orange. 2-Methylpropan-1-ol is branched but primary (its –OH carbon has two hydrogens), so it is oxidised.

QUICK CHECK

Why is propanal distilled off as it forms when propan-1-ol is oxidised?

- So it leaves the oxidising agent before it can be oxidised further to propanoic acid
- Because propanal decomposes if it is heated under reflux
- Because distillation speeds up the oxidation of the alcohol
- So the dichromate(VI) can be reused

An aldehyde is easily oxidised to a carboxylic acid. Propanal boils at 49 °C, well below the alcohol and the acid, because its molecules cannot hydrogen bond to each other, so it can be removed from the hot mixture as soon as it is made.

QUICK CHECK

An IR spectrum has a strong, sharp absorption at 1715 cm^{-1} and no absorption between 2500 and 3600 cm^{-1} apart from sharp peaks near 2950 cm^{-1}. The compound has M_r = 58. What is it?

- propan-1-ol
- propanoic acid
- propanone
- prop-2-en-1-ol

The C=O trough at 1715 cm^{-1} rules out the alcohols, and the absence of any broad O–H trough rules out the acid (and the alcohols). Propanone, CH_3COCH_3, has M_r = 58. Prop-2-en-1-ol has the same M_r but would show a broad alcohol O–H trough and no C=O.

QUICK CHECK

Which gas in the atmosphere does *not* absorb infrared radiation?

- carbon dioxide
- methane
- water vapour
- nitrogen

A bond absorbs IR only if its vibration changes the polarity of the molecule. N≡N joins two identical atoms, so stretching it never creates a dipole. CO_2 is non-polar overall, but its asymmetric stretch and bending vibrations do produce a dipole, so it absorbs.